NAME:ASHOK

REG.NO:192312558

Ex 7: Breadth First Search (BFS)

CODE:

```
from collections import deque
def bfs(graph, start):
  visited = set()
  queue = deque([start])
  visited.add(start)
  order = []
  while queue:
    node = queue.popleft()
    order.append(node)
    for neighbor in graph.get(node, []):
      if neighbor not in visited:
        visited.add(neighbor)
        queue.append(neighbor)
  return order

graph = {
  'A': ['B', 'C'],
  'B': ['D', 'E'],
  'C': ['F'],
  'D': [],
```

```
    'E': ['F'],
    'F': []
}
print("BFS Traversal:", bfs(graph, 'A'))
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-7.py ==============
BFS Traversal: ['A', 'B', 'C', 'D', 'E', 'F']
```